

Received 12 May 2025, accepted 18 May 2025, date of publication 26 May 2025, date of current version 18 June 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3573719

RESEARCH ARTICLE

MFN: Multi-Scale Frequency Feature Fusion Network for Multi-Classification Image Segmentation

JI XIAO¹, LI JIANFANG¹, ZHAO PENG¹, LI XIAOCHEN¹, ZHANG CHENGCHUN¹, ZHENG JUNYI¹, PANG YONGHUI², AND HUANG XIANGSHENG³

¹China Satellite Network Digital Technology Company Ltd., Hebei 071700, China

²Hebei University of Technology, Tianjin 300401, China

³Hebei Key Laboratory of Cognitive Intelligence, Xiong'an Institute of Innovation, Hebei 071700, China

Corresponding authors: Li Jianfang (lijf2@chinasatnet.com.cn) and Huang Xiangsheng (huangxiangsheng@xii.ac.cn)

ABSTRACT Semantic segmentation is a fundamental task in computer vision. In multi-class scenarios, the abundance of categories, feature similarity across classes, class imbalance, and the complexity of feature space modeling pose significant challenges in delineating features and class boundaries. Recent multi-class segmentation methods often suffer from issues such as boundary ambiguity, loss of detail, and inadequate preservation of contextual information. To address these challenges, this paper introduces the Multi-scale Frequency Feature Fusion Network (MFN), a convolutional neural network designed to effectively integrate multi-scale information. First, the Deeplab network architecture is enhanced by refining the ASPP module. Finer-grained atrous convolutions are introduced in its channels to capture multi-scale semantic information, enabling the extraction of richer and more detailed features. The network incorporates a hybrid attention module to enhance the importance of specific spatial points and strengthen feature dependencies between channels. During the feature fusion stage, the FreqFusion module adaptively smooths high-level features to address class inconsistency and boundary misalignment effectively. And a combined loss function, integrating cross-entropy loss and the Dice coefficient, is designed to improve model training and achieve optimal segmentation performance. Extensive experiments conducted on the Cityscapes and Pascal VOC datasets demonstrate that the proposed method achieves superior performance.

INDEX TERMS Semantic segmentation, multi-class, multi-scale, feature fusion.

I. INTRODUCTION

Semantic segmentation is a core task in computer vision that assigns semantic category labels to each pixel in an input image. It serves a critical role in various fields, including autonomous driving, medical image analysis, and mobile applications. Traditional segmentation methods partition an image into several non-overlapping regions based on features such as intensity, color, texture, and shape, ensuring similarity within regions and significant differences between regions. Representative approaches include threshold-based methods, edge detection, region-based segmentation, graph theory, and wavelet transform-based methods.

The associate editor coordinating the review of this manuscript and approving it for publication was Furkan Ahmad¹.

Threshold-based segmentation methods classify pixels by calculating and comparing their intensity against predefined thresholds. For example, Shihua et al. [1] proposed a multi-threshold segmentation approach using line-intercept histograms. Edge-based segmentation methods, such as those by Hu et al. [2] and Wang et al. [3], identify abrupt changes in intensity, color, or texture to detect edges. Operators like Roberts (Mu et al. [4]), Sobel, Prewitt (Bai-Xian et al. [5]), and Laplacian (Yu-Feng et al. [6]) are used to extract edge information and facilitate segmentation. Graph-based segmentation methods relate image segmentation to the minimum cut problem, where specific edges are removed to partition the graph into subgraphs. For example, Felzenszwalb et al. [7] introduced a graph-based segmentation approach utilizing a greedy clustering algorithm.

and a deformable parts model. Wavelet transform-based methods determine edges by detecting modulus maxima points across scales, extracting high-quality edge information for segmentation. For instance, Li et al. [8] used wavelet transforms and an adaptive global thresholding method to extract sharper edge features for segmentation based on image distribution and texture. However, traditional feature extraction methods face significant limitations in addressing the extensive feature set and inter-class similarities present in multi-class images. The advent of deep learning has brought transformative progress to semantic segmentation, achieving remarkable results and addressing the lack of semantic information in traditional segmentation methods. In 2014, Fully Convolutional Networks (FCNs) [9] pioneered the application of deep learning to image segmentation, achieving pixel-level segmentation and establishing a foundation for using deep learning-based neural networks to address segmentation problems. U-Net [10], an enhancement of FCN, achieved more accurate pixel boundary localization by capturing contextual information and employing skip connections. In 2017, PSPNet [11] demonstrated outstanding performance by introducing a pyramid pooling module to capture features from different regions. DeeplabV3+ [12] was introduced, employing spatial pyramid modules to expand the receptive field and enhance model performance through multi-scale information. In 2018, CBAM [13] was proposed, enhancing feature refinement with minimal computational overhead. AutoFocusFormer [14] adopted an adaptive down-sampling process and a local attention module to generate point-based representations for segmentation heads. In 2024, MANet [15], an end-to-end network architecture, introduced multi-scale convolution kernels and a hybrid attention mechanism to improve segmentation accuracy. However, while these methods expand the receptive field and scale, they often neglect fine-grained image features, failing to balance contextual semantic information with detailed features.

To address this issue, this paper proposes a new model called MFN, based on DeepLabv3+. The ASPP module in DeepLabv3+ is improved by introducing the MASA module, which enhances the model's ability to capture fine-grained features while maintaining access to multi-scale information. Additionally, a meticulously designed SCBAM attention module is incorporated to improve the capture of edge information in the features. During the feature fusion stage, the FreqFusion module replaces the conventional bilinear interpolation upsampling method, making the features learnable and adaptively smoothing high-level features while connecting them with low-level features. This enhances the correlation between information at different scales, thereby improving segmentation accuracy. A joint loss function is also designed by combining cross-entropy loss and Dice coefficient loss in a weighted manner, enabling more effective model training and achieving the desired segmentation performance. To summarize, the contributions of the proposed MFN are as follows:

- We proposed the MASA module, which combines large-kernel dilated convolutions with standard convolutions to capture multi-granular features and long-range dependencies. Additionally, the SCBAM attention mechanism is designed to capture edge information, significantly enhancing the model's representational power.
- We integrated the FreqFusion module during the feature fusion stage to address class inconsistency and boundary displacement issues by resampling nearby consistent features to replace inconsistent high-level features. Furthermore, the high-frequency boundary details of low-level features are enhanced to increase high-frequency power, thereby sharpen boundaries.
- Extensive experimental results on Cityscapes and PASCAL VOC 2012 demonstrate that the proposed MFN achieves outstanding performance in multi-class semantic segmentation.

II. RELATED WORK

Semantic scene segmentation is a critical research area in computer vision, widely applied across various tasks. With advancements in imaging technology, the resolution of acquired images continues to improve, leading to a significant increase in image volume and complexity. This trend places higher demands on segmentation algorithms to capture target information comprehensively and improve interpretative efficiency. Traditional segmentation algorithms, such as Support Vector Machines (SVM) [16], Random Forests (RF) [17], and Conditional Random Fields (CRF) [18], exhibit limited capability in extracting target information from high-resolution images, making it difficult to ensure segmentation accuracy. Moreover, these algorithms perform poorly when faced with the demands of real-time image interpretation, as their computational efficiency often falls short of practical requirements. With the rise of neural networks, deep learning methods have demonstrated the ability to automatically learn and extract more robust and discriminative features from images in a data-driven manner. These methods have shown remarkable performance in semantic scene segmentation, surpassing many traditional approaches in terms of both accuracy and efficiency. Notably, the Deeplab series networks, which utilize Convolutional Neural Networks (CNN) as their backbone architecture, have gained widespread recognition and application in semantic scene segmentation due to their use of atrous convolutions and the fusion of deep and shallow features. However, CNNs are generally more adept at extracting local features, such as edges and textures, from images. For more complex image segmentation tasks, relying solely on local features is insufficient; global contextual information and multi-scale features must also be considered. Therefore, image segmentation tasks must not only focus on enhancing the network's ability to perceive contextual information but also emphasize designing new feature fusion strategies

to prevent the loss of feature information across different classes.

A. BRIEF INTRODUCTION OF DEEPLAB

The DeepLab series networks, developed by Google's team, were designed for semantic segmentation tasks. Over time, four versions have been introduced, and the DeepLab networks remain widely used in image semantic segmentation. Their innovative use of atrous convolution and feature sampling fusion techniques continues to be influential, with many researchers proposing optimizations and improvements based on the original DeepLab networks. The development of the DeepLab networks was inspired by PSPNet, a scene parsing network based on a pyramid module that aggregates contextual information across different scales using a pyramid pooling module. Drawing on the pyramid module concept, Chen et al. in DeepLab V1 [19] removed the fully connected layers and the last two pooling layers while introducing atrous convolution into the VGG16 network to enlarge the receptive field. They also proposed the use of Conditional Random Fields (CRF) to improve classification accuracy. In 2017, Google's team introduced DeepLab V2 [20], which utilized atrous convolution and a spatial pyramid pooling module to expand the network's receptive field. In DeepLab V3 [12], the ASPP structure was modified by adding 1×1 convolutions and a global pooling layer. In the subsequent DeepLabv3+, depthwise separable convolutions replaced the original pooling layers, and the Xception model was introduced to deepen the network structure, ensuring that features extracted using atrous convolution were not lost. Additionally, Zhao et al. [11] proposed the Pyramid Pooling Module (PPM), which incorporates information from different scales and subregions. Long et al. [21] proposed a multi-scale feature extraction method for neural networks based on convolutional kernels. Their method introduced an adaptive focus convolution layer consisting of parallel convolution layers with different dilation rates and an attention module to learn the weights of different branches, enabling adaptive adjustment of the receptive field size to extract multi-scale features. Lin et al. [22] employed two parallel encoders to extract information from different scales and utilized a single decoder to integrate the information. Zhang et al. [23] used three parallel CNNs with weighted connections to extract multi-scale information. DenseASPP [24] employed multiple convolutions with varying dilation rates in parallel to extract multi-scale features. In this study, we adopt DeepLabv3 as the base model and explore methods to further enhance its performance to address more complex real-world applications. By incorporating a carefully designed decoder module and feature fusion module, our objective is to enhance segmentation accuracy, particularly in image boundary regions, achieving superior results in various segmentation benchmarks and providing robust technical support for practical drone aerial imagery applications.

B. SEMANTIC SEGMENTATION BASED ON ATTENTION MECHANISM

In addition to extracting local contextual information, some methods also leverage non-local contextual information through attention modules. Peng et al. [25] proposed the Multi-Attention Network (MANet), which applies the self-attention mechanism at each stage of feature extraction and then fuses the extracted feature maps to capture contextual information at different levels. Zhu et al. [26] proposed Bi-level Routing Attention, where each image patch is associated with a positional router. These routers assign the image patches to upper and lower-level routers based on specific rules. The upper-level router captures global context information, while the lower-level router focuses on local region details. This approach allows the model to capture both global and local features, thereby enhancing performance in visual tasks. Lv et al. [27] proposed a new Mask-CSP-Attention Coupling Network (MCAN) to enhance detail detection and improve accuracy in single-sample detection. The results demonstrated that this network achieved high-precision segmentation across various environments. Zheng et al. [28] improved the structure of the High-Resolution Network (HR-Net) algorithm, making it more suitable for forest extraction in remote sensing images. However, the above network did not focus on the ability to extract multi-scale features. Liu et al. [29] proposed a Multi-Scale Channel Spatial Attention Network (MSCSA-Net), which improved the overall accuracy of tree segmentation. However, this network only utilized convolutional layers to construct channel and spatial attention modules and did not adopt the Transformer encoder-decoder structure, limiting its ability to extract global features. Neha et al. [30] proposed using residual connections between convolutional layers and integrating Multi-layer Feature Fusion (MFF) and Cross-Channel Attention (CCA) in encoder blocks. They also enhanced skip connections with additional information generated by MFF and CCA. Swin-Unet [31] introduced the SENt (Squeeze-and-Excitation) module, which uses a channel attention mechanism to weight feature maps, improving the model's ability to capture detailed and spatial information while further enhancing segmentation performance. The SENt module recalibrates channel importance through global average pooling and two fully connected layers, allowing the network to adaptively emphasize useful features and suppress irrelevant ones. SE-RegUNet [32]: This model enhances performance by introducing a RegNet encoder and squeeze-excitation blocks. This structure leverages the strong feature extraction capabilities of the RegNet encoder while using the SE module to enhance feature representation. AS-Unet++ [33]: Built upon the traditional Unet, AS-Unet++ adds ASPP and SE modules to expand the receptive field and enhance the capture of important features. By stacking the feature extraction layers of AS-Unet, AS-Unet++ reduces the training load of multiple AS-Unets. Compared to Unet, AS-Unet++ reduces the number of training parameters while maintaining efficient

feature fusion and representation. Xie et al. [34] proposed an extended U-Net model named SEA-Net, which utilizes compressed and expanded models along with attention mechanisms. SEA-Net enhances the model's ability to refine feature processing, particularly when handling images with complex structures and details, by combining channel and spatial attention. Existing research focuses on aggregating contextual information from within a single image, while overlooking the importance of pixel representations from corresponding classes outside the input image. To address this contradiction, Jin et al. [35] proposed enhancing each pixel's representation by aggregating dataset-level representations, thereby extracting contextual information beyond a single image to further enrich pixel representations.

C. MULTI-SCALE FEATURE FUSION METHODS

Multi-scale refers to sampling an image at different granularities, allowing the observation of various features at different scales to meet task requirements. Generally, finer and denser sampling reveals more detailed features, while coarser and sparser sampling captures the overall trend. MTCNN [36] fuses images at different scales by resizing them to various dimensions to capture information from images of different sizes. SPPNet [37] proposed the Spatial Pyramid Pooling Network, which extracts features using blocks of varying sizes and applies pooling operations to these extracted features, thereby achieving multi-scale feature extraction. The introduction of dilated convolution [38] alleviates the loss of detail information caused by pooling operations in the network. Building on this, ASPP resamples convolutional features extracted at a single scale to accurately classify regions of different scales. FPN [22] can be divided into shallow and deep networks. The shallow network is responsible for extracting low-level features, such as object edges, while the deep network constructs high-level semantic information. By using this approach, high-level semantic information from the deep network is fused into the shallower network, guiding the shallow network in its recognition tasks. UNet improves upon FPN by using skip connections to link deep and shallow layers with features, facilitating better gradient information propagation during training. HRNet [39] uses multiple parallel branches at different resolutions and continuously exchanges information between branches, achieving strong semantic information and precise spatial positioning. SAFA [40] proposed a scale-adaptive feature aggregation method, allowing the network to adaptively select sub-grids of different scales for each sample. CEDNet [41] uses a cascaded feature fusion method to perform multi-scale feature fusion at the decoder stage, integrating high-level features early on to guide the fusion of lower-level features later. CFP [42] utilizes a concentrated feature pyramid structure, improving the pyramid connection to globally optimize features while focusing more on small targets in corners. The extraction and fusion of multi-scale features allow the network model to integrate information from different scales. This enables

the model to better capture objects, details, and contextual features in an image, enhancing its classification of objects of different sizes and improving performance across various tasks. The MLCA2F module [43] has been developed for the purpose of conducting detailed contextual segmentation, whilst the BG-3DM2F module [44] incorporates a chained 3D architecture with a bidirectional recurrent fusion unit. The dermatology CAD strategy involves the fusion of multi-layer CNN features under the framework of transfer learning [45]. The following papers constituted the basis of the research: CNN's use of Gaussian pyramids with region proposals for efficient multi-scale analysis [46]; CafeNet's combination of multi-scale context aggregation and inter-level foreground guidance [47]; and OATC's pipeline for simultaneous multi-target ROI extraction in wrist X-rays [48]. In summary, the aforementioned studies demonstrate the adaptability of multi-scale fusion techniques ranging from handcrafted region selection to deep attention modules to a variety of imaging problems, each employing innovative network designs and fusion schemes.

III. METHODOLOGY

A. NETWORK ARCHITECTURE

As shown in Figure 1, the proposed MFN consists of four components: the Feature Extraction Module (FE), the Multi-Scale Attention Module with Pyramid Architecture (MASA), the Feature Fusion Module (FreFusion), and the final segmentation head module (CLS). In the field of machine learning, the FE module is a crucial component in the development of deep neural networks. It is in this module that the choice of network architecture, such as Xception, ResNet, and MobileNet, is of paramount importance. In selecting MobileNet for the FE module, the objective is to ensure that the model achieves optimal accuracy with minimal parameters, thereby meeting the requirement for a lightweight model. Given an input image $I \in \mathbb{R}^{3 \times H \times W}$, the image is first input to the FE module, where convolution is performed for feature extraction, ultimately outputting two feature maps. One part is the feature maps output by the last convolutional layer, and the other part is the feature maps of the intermediate low-level features, as shown below:

$$F_S, F_D = f_{FE}(I) \quad (1)$$

Subsequently, the deep features F_D are fed into the MASA module for the extraction of multi-scale information, denoted as f_{MASA} . This process can be represented by the following formula:

$$F_M = f_{MASA}(F_D) \quad (2)$$

In the equation, F_M represents the high-level features used for final reconstruction. By employing the Feature Fusion Module FreqFusion, F_M is integrated with the low-level features F_S output by the FE module. The final output is achieved by combining these high-level features with the low-level features, as expressed by the following formula:

$$F_r = f_{Fre}(F_M, F_S) \quad (3)$$

Ultimately, the target image size is restored through the CLS structure, and a convolutional module is used to segment the target areas within the image, resulting in the predicted image $I_{pre} \in \mathbb{R}^{3 \times H \times W}$, where N denotes the number of segmented classes.

$$I_{pre} = f_{CLS}(F_r) \quad (4)$$

In the translation: where $f_{CLS}(\cdot)$ represents the segmentation module implemented by a 3×3 convolution, upsampling, and another 3×3 convolution, to achieve image segmentation prediction.

B. SELF-SUPERVISED PRETRAINING

To reduce reliance on large-scale labeled data, we pretrain the CNN backbone using a contrastive learning framework. Given an unlabeled image I , we generate two augmented views I_1 and I_2 via random cropping and color jittering. The model is trained to maximize the similarity between feature embeddings of I_1 and I_2 while minimizing similarity with negative samples. The loss function is defined as:

$$\mathcal{L}_{self} = -\log \frac{\exp(\text{sim}(f(I_1), f(I_2))/\tau)}{\sum_{k=1}^K \exp(\text{sim}(f(I_1), f(I_k))/\tau)}, \quad (5)$$

where τ is a temperature parameter, and K is the number of negative pairs. The results of comparative trials conducted using different training strategies are presented in Table 1.

TABLE 1. Performance of different pre-training strategies.

Pretraining Strategy	mIoU (Pascal VOC)
ImageNet Pretraining	88.36
Self-Supervised	83.12
No Pretraining	78.30

C. LOSS FUNCTION

The proposed joint loss function combines cross-entropy loss (L_{CE}) and Dice coefficient loss (L_{Dice}) to balance class imbalance and boundary optimization. The calculation is performed as follows:

1) Cross-Entropy Loss:

$$L_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(p_{i,c}), \quad (6)$$

where N is the number of pixels, C is the number of classes, $y_{i,c}$ is the ground truth label (0 or 1), and $p_{i,c}$ is the predicted probability of pixel i belonging to class c .

2) Dice Coefficient Loss:

$$L_{Dice} = 1 - \frac{2 \sum_{i=1}^N y_i p_i + \epsilon}{\sum_{i=1}^N y_i + \sum_{i=1}^N p_i + \epsilon}, \quad (7)$$

where ϵ is a smoothing factor to avoid division by zero.

3) Joint Loss:

$$L = \lambda L_{CE} + (1 - \lambda) L_{Dice}, \quad (8)$$

where $\lambda = 0.7$ is empirically determined to balance convergence speed and segmentation accuracy. The weights of the loss function are compared to the test as shown in the Table 2.

TABLE 2. A comparative analysis of the weights of loss functions.

λ	0	0.3	0.5	0.7	0.9	1
mIoU(%)	86.56	87.01	87.45	88.36	88.03	86.87

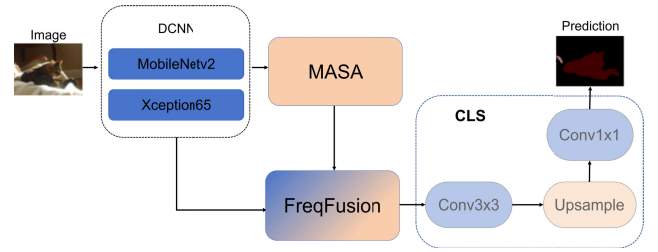


FIGURE 1. Overview of MFN.

D. MULTI-SCALE DILATED PYRAMID ATTENTION (MASA)

Inspired by PSPNet and DeepLab, we improve the structure of the original Pyramid Pooling Module (ASPP), resulting in the MASA module, as shown in Figure 2. The MASA module consists of two substructures: the Multi-ASPP Module (MASPP) and the Skip-CBAM Module (SCBAM). The MASPP module enhances the pyramid structure by incorporating convolution modules with varying dilation rates and performing skip connections between different dilation rates. This approach not only captures features across multiple scales but also strengthens the relationships between different channels, extracting more effective features and class boundaries. The SCBAM module combines channel attention and spatial attention to form a hybrid attention mechanism, while simultaneously performing skip connections on the original input. This ensures that feature information is fused across both the channel and spatial dimensions without losing the original feature's integrity, thus preserving more deep feature information. Given an input feature, the entire process of MASA is as follows:

$$X = \text{MASPP}(X),$$

$$X = f_{1 \times 1}(X),$$

$$X = \text{SCBAM}(X),$$

$$\text{SCBAM}(X) = \text{Concat}(\text{SAM}(\text{CAM}(X)), X) \quad (9)$$

where MASPP and SCBAM represent the previously introduced MASPP and SCBAM modules, respectively, and convolution is used to change the dimensionality.

E. FREQUENCY-AWARE FEATURE FUSION(FreqFusion)

FreqFusion [49] is a method used to improve feature fusion performance in computer vision tasks. It adaptively smooths

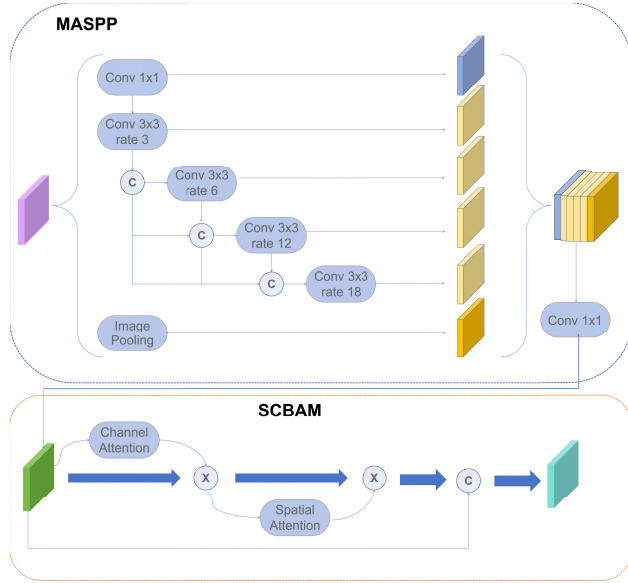


FIGURE 2. Structure of MASA.

high-level features using spatially varying low-pass filters and replaces inconsistent features in high-level features by resampling nearby features of the same class, thereby addressing issues of class inconsistency and boundary displacement. This method leverages frequency characteristics to enhance both low-resolution and high-resolution features. It is implemented through the following three key components. The structure of the three generator components is shown in Figure 4. And Table 3 presents a comprehensive ablation study, meticulously assessing the individual contributions of the Adaptive Low-Pass Filter (ALPF), Adaptive High-Pass Filter (AHPF), and offset generators in the final fusion process of FreqFusion.

- (1) **Adaptive Low-Pass Filter (ALPF) Generator:** This generator predicts spatially varying low-pass filters that reduce class inconsistency within objects during the upsampling process and smooth the features. In this way, it reduces differences between pixel values and minimizes feature inconsistencies.
- (2) **Offset Generator:** This generator predicts offsets for resampling feature pixels and replaces inconsistent features in high-level features. This helps maintain class consistency during feature fusion.
- (3) **Adaptive High-Pass Filter (AHPF) Generator:** This generator predicts high-pass filters that enhance high-frequency boundary details in low-level features. This helps increase the high-frequency power of features and sharpens the boundaries.

To enhance the model's segmentation ability for complex categories and backgrounds, a residual connection between high-frequency and low-frequency features is added to improve the model's ability to represent complex data. The overall structure of the FreqFusion is shown in Figure 3. The module uses low-frequency and high-frequency features as input, with the low-frequency features guiding the

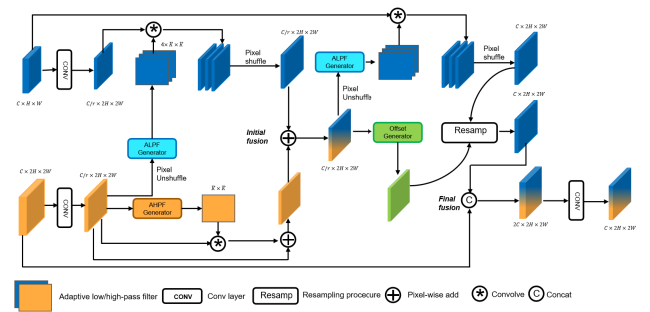


FIGURE 3. The overview of FreqFusion. Pixel unshuffle involves resizing the spatial dimensions of the feature by half and expanding the channel by a factor of, dividing them into 4 groups, such as from $C \times 2H \times 2W$ to $4C \times H \times W$. Pixel shuffle is the reverse operation, transitioning from $4C \times 2H \times 2W$ to $C \times 2H \times 2W$. The Adaptive Low-Pass Filter (ALPF) generator and Adaptive High-Pass in the initial fusion share the same parameters as those in the final fusion.

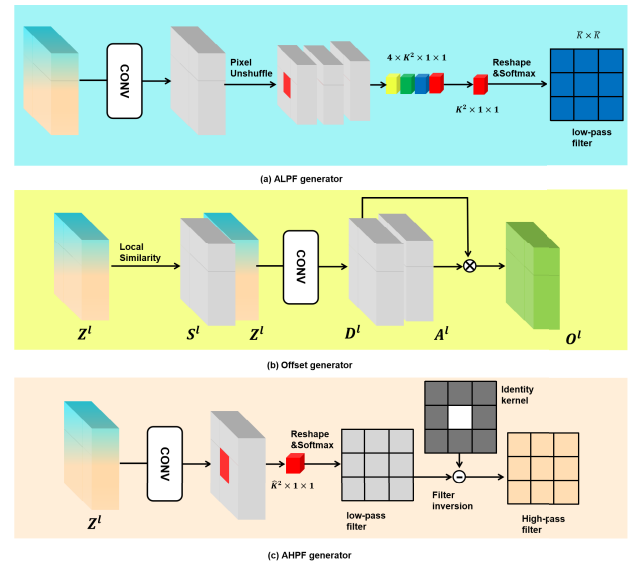


FIGURE 4. The overview of generators in FreqFusion. $\otimes$ represents element-wise multiplication, and $\ominus$ represents subtraction.

TABLE 3. Ablation study for the final fusion of FreqFusion. Results are reported on ADE20K validation set.

Deeplabv3+	ALPF generator	AHPF generator	Offset generator	Params (M)	FLOPs(G)	mIoU(%)
Standard feature fusion	×	×	×	4.26	6.59	41.1
FreqFusion	✓	✓	✓	4.42	6.91	42.0(+0.9)
	✓	✓	✓	4.43	6.97	42.9(+1.8)
	✓	✓	✓	4.59	7.14	43.5(+2.4)

high-frequency features to correct deviations in internal features. The FreqFusion module consists of two parts: initial fusion and final fusion. Some previous works still use simple linear interpolation methods, which inevitably lead to class-internal feature inconsistencies and feature similarity issues, the pseudocode for this module is presented in Algorithm 1.

IV. EXPERIMENTS

A. DATASETS AND METRICS

We conduct experiments with MFN on two datasets: Cityscapes Cordts et al. [50], and PASCAL VOC 2012

Algorithm 1 Algorithm of FreqFusion

Input: F_M (High-level coarse feature (low resolution)),
 F_S (Low-level fine feature (high resolution))
Output: F_r (Fused high-resolution feature)

- 1: **Initial Fusion**
- 2: $Z \leftarrow \text{Conv}_{1 \times 1}(\text{UpSample}(F_M)) + \text{Conv}_{1 \times 1}(F_S)$ {Feature compression}
- 3: **Adaptive Low-Pass Filter (ALPF)**
- 4: $\hat{V}_{\text{low}} \leftarrow \text{Conv}_{3 \times 3}(Z)$ {Generate filter weights}
- 5: $\hat{W}_{\text{low}} \leftarrow \text{Softmax}(\hat{V}_{\text{low}})$ {Spatially-variant LPF}
- 6: $Y_{\text{filtered}} \leftarrow \hat{W}_{\text{low}} * F_M$ {Apply filtering}
- 7: $Y_{\text{upsampled}} \leftarrow \text{PixelShuffle}(Y_{\text{filtered}})$ {Sub-pixel upsampling}
- 8: **Offset Generator**
- 9: $S \leftarrow \text{ComputeLocalSimilarity}(Z)$ {Compute local cosine similarity}
- 10: $D \leftarrow \text{Conv}_{3 \times 3}(\text{Concat}(Z, S))$ {Offset direction}
- 11: $A \leftarrow \text{Sigmoid}(\text{Conv}_{3 \times 3}(\text{Concat}(Z, S)))$ {Offset magnitude}
- 12: $\text{offsets} \leftarrow D \odot A$ {Dynamic offsets}
- 13: $Y_{\text{resampled}} \leftarrow \text{Resample}(Y_{\text{upsampled}}, \text{offsets})$ {Feature resampling}
- 14: **Adaptive High-Pass Filter (AHPF)**
- 15: $\hat{V}_{\text{high}} \leftarrow \text{Conv}_{3 \times 3}(Z)$
- 16: $\hat{W}_{\text{high}} \leftarrow \mathbf{E} - \text{Softmax}(\hat{V}_{\text{high}})$ { $\mathbf{E}$: Identity kernel}
- 17: $X_{\text{enhanced}} \leftarrow X_{\text{low}} + \hat{W}_{\text{high}} * X_{\text{low}}$ {Residual high-frequency enhancement}
- 18: **Final Fusion**
- 19: $Y_{\text{fused}} \leftarrow Y_{\text{resampled}} + X_{\text{enhanced}}$
- 20: $F_r \leftarrow \text{CONCAT}(Y_{\text{fused}}, X_{\text{low}})$
- 21: **return** F_r

Everingham et al. [51], to demonstrate the effectiveness of our method. We first pre-train the CNN backbone on ImageNet Deng et al. [52], and then fine-tune it on semantic segmentation datasets. Additionally, we apply two standard evaluation metrics—mean intersection over union (mIoU) and F1-score (F1) to assess the quality of image segmentation. mIoU is a commonly used evaluation metric for semantic segmentation models. It measures the model's performance by calculating the ratio between the intersection and union of predicted results and ground truth labels. The formula for mIoU is as follows:

$$mIoU = \frac{1}{k+1} \sum_{i=0}^k \frac{TP}{FN + FP + TP} \quad (10)$$

where TP denotes True Positives, FP denotes False Positives, and FN denotes False Negatives. mIoU considers the pixel-level prediction accuracy of the model and provides a fair evaluation of the model's performance on targets of different categories and sizes.

The F1-score is the harmonic mean of precision and recall, used to provide a comprehensive evaluation of model performance. The formula for calculating F1-score

is as follows:

$$Recall = \frac{TP}{TP + FN} \quad (11)$$

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

$$F1 = \frac{2 \times (Precision \times Recall)}{Precision + Recall} \quad (13)$$

The advantage of the F1-score lies in its consideration of both precision and recall, providing a more accurate performance evaluation of the model in imbalanced classification problems. In semantic segmentation tasks, the F1-score helps us understand the model's overall performance across different categories.

B. TRAINING ENVIRONMENT AND CONFIGURATION

During the training phase, data augmentation was performed through horizontal flipping and random rotations of 90°, 180°, and 270° to enhance the training effect. For training from scratch and fine-tuning stages, the $\{\text{image size}, \text{batch size}\}$ was set to $\{512 \times 512, 32\}$. A combined loss was used to distinguish the accuracy of pixel-level classification quality. All models were trained using the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.99$. The learning rate was initialized at 5×10^{-4} . During retraining, a cosine annealing learning rate schedule was used for 1600 K iterations, while for fine-tuning, it was set to 1×10^{-4} for 800 K iterations. All experiments were conducted using the PyTorch framework on 4 Nvidia RTX 3090 GPUs.

C. COMPARISON WITH STATE-OF-THE-ART METHODS**1) RESULTS ON CITYSCAPES**

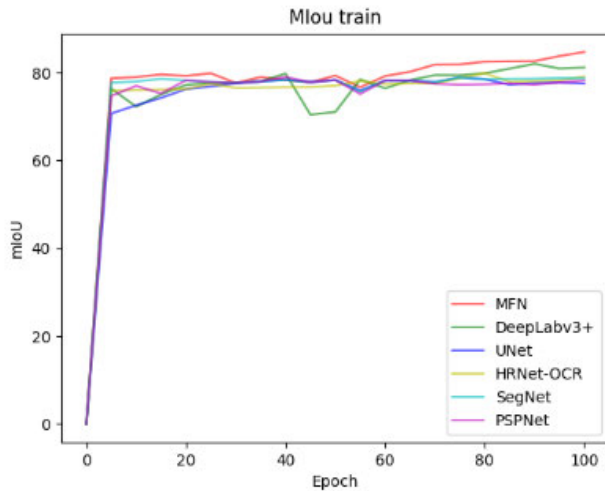
The corresponding results for Cityscapes are shown in Table 4. Our MFN has achieved excellent performance to a great extent. For instance, our MFN has attained an 83.12 mIoU and an 91.00 F1-score. Compared to PSPNet and DeepLabV3+, our method exhibits better accuracy. Our MFN is faster across all input resolutions and yields superior mIoU results compared to all other methods. Furthermore, when compared to SegNet and HRNet-OCR, our MFN also achieves a better trade-off. Furthermore, a comparison with the contemporary transformer-based methods Swin-UNet and Segformer was conducted, and it was demonstrated that our method exhibits superior inference speed. Figure 5 shows the plot of mIoU on the Cityscapes dataset based on pre-training using ImageNet.

2) RESULTS ON PASCAL VOC 2012

The corresponding results for Pascal VOC 2012 are shown in Table 5. Our MFN has achieved the best mIoU and F1-score. For example, we have achieved 88.36 mIoU and 91.33 F1-score on VOC. Considering the diverse images and various semantic categories in Pascal VOC 2012 Everingham et al. [51], this outstanding result further demonstrates the generalization capability of our MFN.

TABLE 4. Comparisons with other state-of-the-art segmentation methods on Cityscapes.

Method	mIoU (%)	F1-score(%)	FLOPs(G)	Params(M)	FPS
SegNet	78.85	88.80	24.3	29.5	36.5
U-Net	78.23	87.96	341.29	29.64	41.1
PSPNet	78.15	88.10	118.45	46.71	35.9
DeepLabv3+	81.12	89.96	400	40.8	37.83
HRNet-OCR	78.30	88.20	747.3	65.9	57.93
Swin-U-Net	75.20	80.00	210	62	15
SegFormer	83.50	91.80	120	84	18
MFN(ours)	83.12	91.00	376.9	55.2	38.05

**FIGURE 5.** mIoU curve based on ImageNet pretraining.**TABLE 5.** Comparisons with other state-of-the-art segmentation methods on Pascal VOC 2012.

Method	mIoU (%)	F1-score(%)	FLOPs(G)	Params(M)	FPS
SegNet	82.48	89.20	24.3	29.5	36.5
U-Net	79.38	89.03	341.29	29.64	41.1
PSPNet	85.41	88.52	118.45	46.71	35.9
DeepLabv3+	84.56	90.23	400	40.8	37.83
HRNet-OCR	81.0	88.61	747.3	65.9	57.93
Swin-U-Net	78.50	85.00	180	62.0	15
SegFormer	84.00	90.20	78	84	22
MFN(ours)	88.36	91.33	376.9	55.2	38.05

D. ABLATION STUDY

To further validate the effectiveness of the proposed method, ablation experiments were conducted on the validation set, comparing network architectures with and without the MASA module and the FreqFusion module. The experimental results are shown in Table 6. In the ablation experiments, the joint loss function proposed in this paper was used as the model's loss function to compare the overall performance of the two modules. The results show that the MFN with the MASA module achieved improvements in both the mean Intersection over Union (mIoU) and F1-score on the two datasets, with a maximum increase of 5.23% in mIoU and 0.36% in F1-score. Similarly, the MFN with the FreqFusion module also showed improvements in both mIoU and F1-score on the two datasets, with a maximum increase of 2.48% in mIoU and 1.64% in F1-score. The overall highest performance of the model was: mIoU: 88.36%, F1-score: 91.33%. This demonstrates that both the MASA and FreqFusion modules significantly

TABLE 6. Validation set ablation experimental evaluation.

Method/Datasets			Cityscapes		Pascal VOC 2012	
DeepLabv3+	MASA	Freqfusion	mIoU	F1-score	mIoU	F1-score
✓			81.12	89.96	84.36	90.23
✓		✓	82.2	90.32	85.35	90.2
✓	✓		82.6	91.45	86.42	91.87
✓	✓	✓	83.12	91	88.36	91.33

enhance the model's segmentation performance by extracting and integrating multi-scale feature information at different sizes.

V. VISUALIZATION RESULTS

Figure 6 presents the visualization results of our Multi-scale Frequency Feature Fusion Network (MFN) on the Cityscapes validation set. When compared to other state-of-the-art models such as PSPNet and DeepLabv3+, our MFN demonstrates superior performance in several key aspects.

On one hand, for large-area categories like roads, sidewalks, and large trucks, MFN provides more accurate and complete segmentation results. It effectively captures the overall structure and boundaries of these extensive regions, ensuring that they are distinctly and correctly segmented from the surrounding environment. This is crucial for applications such as autonomous driving, where precise identification of drivable areas is essential for safe navigation.

On the other hand, MFN excels in delivering more accurate boundaries for small or thin objects. Objects such as utility poles, traffic lights, traffic signs, and cars often pose challenges for segmentation models due to their relatively small size and thin structures. However, MFN manages to delineate these objects with remarkable precision. It successfully captures the fine details and edges of these objects, ensuring that they are accurately segmented and not lost or merged with the background or other objects.

This dual capability of MFN to handle both large-area categories and small or thin objects effectively demonstrates its robustness and versatility. It showcases the model's ability to extract high-quality long-range context, which is essential for understanding the overall scene and relationships between objects. At the same time, MFN preserves fine details, allowing it to accurately segment even the smallest and most intricate objects within the scene. The visualization results on the Cityscapes validation set highlight the strengths of our MFN in semantic segmentation tasks. Its ability to balance the extraction of global context with the preservation of local details sets it apart from other models and underscores its potential for a wide range of real-world applications.

VI. LIMITATIONS AND FUTURE WORK

A. LIMITATIONS

Challenges in Small Object Segmentation While the proposed MFN shows strong performance on large area categories such as roads and vehicles, it exhibits limitations in accurately segmenting small or elongated objects in complex urban scenes. To illustrate this point, as demonstrated in 7, segmentation may be prone to incomplete segmentation

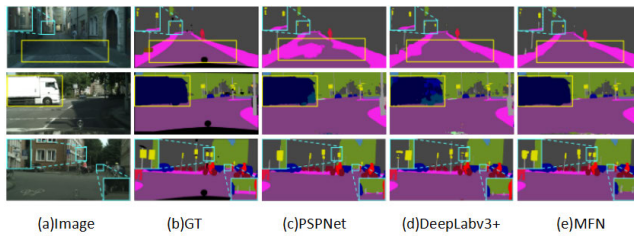


FIGURE 6. Visualization results on Cityscapes validation set. Compared with PSPNet and DeepLabv3+, MFN generates masks with finer details as highlighted in the light blue box and more accurate large-area predictions, as highlighted in the yellow box.

or blurring of boundaries in small structures such as street light poles or thin traffic signs in urban landscape datasets. The reason for this phenomenon is the insufficient resolution of high-level features and the limited sensitivity of the model to low-frequency spatial details during feature fusion.

Performance Degradation in Extreme Class Imbalance

The model exhibits diminished robustness when confronted with datasets characterised by severe class imbalance. For instance, the “sofa” category in Pascal VOC 2012 (which contains a significantly smaller number of training samples in comparison to prevalent categories such as “person” or “car”) attains lower precision and recall values. This phenomenon can be attributed to the inability of the current loss function to dynamically adjust weights for fewer classes, resulting in biased learning towards majority classes.

Limitations of Pretraining Dependency While MFN achieves competitive performance with self-supervised pre-training, its generalization ability on highly domain-specific tasks (e.g., rare disease diagnosis) remains constrained when neither labeled nor unlabeled pretraining data is available. Future work will explore few-shot learning paradigms and domain-invariant feature learning to address this challenge.

B. FUTURE DIRECTIONS

Enhance small object segmentation Preservation of high-resolution features A dedicated high-resolution branch must be included in the decoder with the aim of preserving fine-grained spatial details. This should be complemented by an attention mechanism to enhance the detection of small object features.

Frequency-aware enhancement It is recommended to integrate adversarial training strategies in the data enhancement process in order to improve the model’s sensitivity to fine-grained structures, where the synthesis of high-frequency details (e.g., thin edges) is a particular focus.

The concept of dynamic receptive field tuning is a theoretical framework that aims to explain the process of adapting to external stimuli in a dynamic and receptive manner. The MASA module has to be tuned in order to design meander rates that can be adapted to objects of different sizes, thus ensuring the capture of multi-scale contextual information.

The problem of category imbalance will be addressed in the future by investigating the concept of adaptive loss

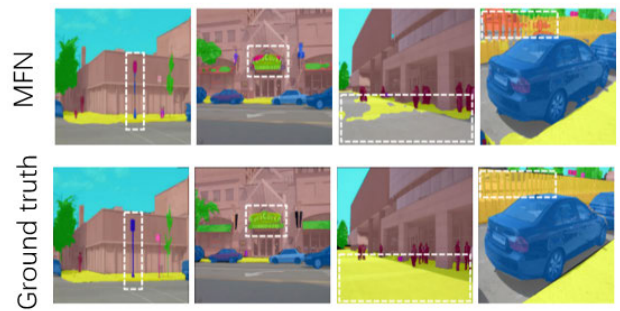


FIGURE 7. Visualization of an incorrectly segmented scene, with a white border showing the region of incorrect segmentation.

design. The existing weighted loss should be replaced by Class-Adaptive Focal Loss, which aims to penalize misclassification in a small number of categories more severely.

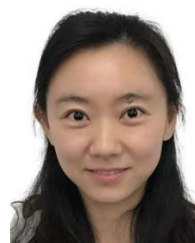
VII. CONCLUSION

In this paper, we propose a novel end-to-end pixel-level segmentation network, which improves segmentation accuracy by integrating a multi-scale dilated convolution pyramid module, and employs the designed SCBAM attention mechanism to enhance segmentation accuracy without increasing the computational cost of the model. We also introduce a new joint loss function that enables faster and more stable parameter updates. The effectiveness and advantages of our MFN are validated on the Cityscapes and VOC datasets, demonstrating higher segmentation accuracy in multi-class tasks, outperforming other methods.

REFERENCES

- [1] S. Wu, Y. Wu, and J. Zhou, “Multi-level thresholding for remote sensing image of urban area based on line intercept histogram,” *CAAI Trans. Intell. Syst.*, vol. 10, Jul. 2018.
- [2] X. Hu, J. Shen, J. Shan, and L. Pan, “Local edge distributions for detection of salient structure textures and objects,” *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 3, pp. 466–470, May 2013.
- [3] J. Wang, X. Yang, X. Qin, X. Ye, and Q. Qin, “An efficient approach for automatic rectangular building extraction from very high resolution optical satellite imagery,” *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 3, pp. 487–491, Mar. 2015.
- [4] K. Mu, X. U. Qinggong, and W. Baoshu, “A Roberts’ adaptive edge detection method,” *J. Xi’an Jiaotong Univ.*, vol. 42, no. 10, pp. 1240–1244, 2008.
- [5] B.-X. Zou, Z. Ran, and M. Jun, “Improved Prewitt method for image edge detection,” *Microelectron. Comput.*, vol. 30, no. 5, pp. 23–26, 2013.
- [6] Y.-F. Gui and J.-P. Wu, “An image edge detection method base on Laplacian operator and degree of gray relation,” *J. Shantou Univ., Natural Sci. Ed.*, vol. 26, no. 2, pp. 69–73, 2011.
- [7] P. F. Felzenszwalb and D. P. Huttenlocher, “Efficient graph-based image segmentation,” *Int. J. Comput. Vis.*, vol. 59, no. 2, pp. 167–181, Sep. 2004.
- [8] L. Li, W. Chai, and S. Mei, “Segmentation of remote sensing images based on adaptive global threshold and fused markers,” *Trans. Chin. Soc. Agricult. Machinery*, vol. 44, no. 7, pp. 222–228, 2013.
- [9] E. Shelhamer, J. Long, and T. Darrell, “Fully convolutional networks for semantic segmentation,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.
- [10] O. Ronneberger, P. Fischer, and T. Brox, “U-net: Convolutional networks for biomedical image segmentation,” in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.
- [11] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, “Pyramid scene parsing network,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.

- [12] L. C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with Atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 801–818.
- [13] S. Woo, J. Park, J. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jan. 2018, pp. 3–19.
- [14] C. Ziwen, K. Patnaik, S. Zhai, A. Wan, Z. Ren, A. Schwing, A. Colburn, and L. Fuxin, "AutoFocusFormer: Image segmentation off the grid," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 18227–18236.
- [15] P. He, L. Jiao, R. Shang, S. Wang, X. Liu, D. Quan, K. Yang, and D. Zhao, "MANet: Multi-scale aware-relation network for semantic segmentation in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.
- [16] Y. Guo, X. Jia, and D. Paull, "Effective sequential classifier training for SVM-based multitemporal remote sensing image classification," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 3036–3048, Jun. 2018.
- [17] M. Pal, "Random forest classifier for remote sensing classification," *Int. J. Remote Sens.*, vol. 26, no. 1, pp. 217–222, Jan. 2005.
- [18] P. Krähenbühl and V. Koltun, "Efficient inference in fully connected CRFs with Gaussian edge potentials," in *Proc. Adv. Neural Inf. Process. Syst.*, 2011, pp. 1–11.
- [19] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," 2014, *arXiv:1412.7062*.
- [20] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, Atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
- [21] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
- [22] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.
- [23] D. Zhang, H. Zhang, J. Tang, M. Wang, X. Hua, and Q. Sun, "Feature pyramid transformer," in *Proc. 16th Eur. Conf. Comput. Vis.*, 2020, pp. 323–339.
- [24] Q. Chen, Y. Wang, T. Yang, X. Zhang, J. Cheng, and J. Sun, "You only look one-level feature," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13034–13043.
- [25] C. Peng, K. Zhang, Y. Ma, and J. Ma, "Cross fusion net: A fast semantic segmentation network for small-scale semantic information capturing in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601313.
- [26] L. Zhu, X. Wang, Z. Ke, W. Zhang, and R. Lau, "BiFormer: Vision transformer with bi-level routing attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10323–10333.
- [27] L. Lv, X. Li, F. Mao, L. Zhou, J. Xuan, Y. Zhao, J. Yu, M. Song, L. Huang, and H. Du, "A deep learning network for individual tree segmentation in UAV images with a coupled CSPNet and attention mechanism," *Remote Sens.*, vol. 15, no. 18, p. 4420, Sep. 2023.
- [28] Z. Zheng, S. Yu, and S. Jiang, "A domain adaptation method for land use classification based on improved HR-net," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4400911.
- [29] K.-H. Liu and B.-Y. Lin, "MSCSA-net: Multi-scale channel spatial attention network for semantic segmentation of remote sensing images," *Appl. Sci.*, vol. 13, no. 17, p. 9491, Aug. 2023.
- [30] F. Neha and A. K. Bansal, "Multi-layer feature fusion with cross-channel attention-based U-Net for kidney tumor segmentation," 2024, *arXiv:2410.15472*.
- [31] Y. Yang, C. Wang, Q. Zhao, G. Li, and H. Zang, "Se-Swin UNet for image segmentation of major maize foliar diseases," *Engenharia Agrícola*, vol. 44, May 2024, Art. no. e20230097.
- [32] S.-S. Chang, C.-T. Lin, W.-C. Wang, K.-C. Hsu, Y.-L. Wu, C.-H. Liu, and Y. C. Fann, "Optimizing ensemble U-Net architectures for robust coronary vessel segmentation in angiographic images," *Sci. Rep.*, vol. 14, no. 1, p. 6640, Mar. 2024.
- [33] G. Nan, H. Li, H. Du, Z. Liu, M. Wang, and S. Xu, "A semantic segmentation method based on AS-UNet++ for power remote sensing of images," *Sensors*, vol. 24, no. 1, p. 269, Jan. 2024.
- [34] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5987–5995.
- [35] Z. Jin, T. Gong, D. Yu, Q. Chu, J. Wang, C. Wang, and J. Shao, "Mining contextual information beyond image for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7211–7221.
- [36] K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, "Joint face detection and alignment using multitask cascaded convolutional networks," *IEEE Signal Process. Lett.*, vol. 23, no. 10, pp. 1499–1503, Oct. 2016.
- [37] Q. Xu, W. Kuang, Z. Zhang, X. Bao, H. Chen, and W. Duan, "SPPNet: A single-point prompt network for nuclei image segmentation," in *Proc. Int. Workshop Mach. Learn. Med. Imag.*, 2023, pp. 227–236.
- [38] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.
- [39] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5686–5696.
- [40] Z. Huang, A. Huang, X. Hu, C. Hu, J. Xu, and S. Zhou, "Scale-adaptive feature aggregation for efficient space-time video super-resolution," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 4228–4239.
- [41] G. Zhang, Z. Li, C. Tang, J. Li, and X. Hu, "CEDNet: A cascade encoder-decoder network for dense prediction," *Pattern Recognit.*, vol. 158, Feb. 2025, Art. no. 111072.
- [42] Y. Quan, D. Zhang, L. Zhang, and J. Tang, "Centralized feature pyramid for object detection," *IEEE Trans. Image Process.*, vol. 32, pp. 4341–4354, 2023.
- [43] I. Bakkouri and K. Afdel, "MLCA2F: Multi-level context attentional feature fusion for COVID-19 lesion segmentation from CT scans," *Signal, Image Video Process.*, vol. 17, no. 4, pp. 1181–1188, Jun. 2023, doi: [10.1007/s11760-022-02325-w](https://doi.org/10.1007/s11760-022-02325-w).
- [44] I. Bakkouri, K. Afdel, J. Benois-Pineau, and G. C. F. T. A. Initiative, "BG-3DM2F: Bidirectional gated 3D multi-scale feature fusion for Alzheimer's disease diagnosis," *Multimedia Tools Appl.*, vol. 81, no. 8, pp. 10743–10776, Mar. 2022, doi: [10.1007/s11042-022-12242-2](https://doi.org/10.1007/s11042-022-12242-2).
- [45] I. Bakkouri and K. Afdel, "Computer-aided diagnosis (CAD) system based on multi-layer feature fusion network for skin lesion recognition in dermoscopy images," *Multimedia Tools Appl.*, vol. 79, nos. 29–30, pp. 20483–20518, Aug. 2020, doi: [10.1007/s11042-019-07988-1](https://doi.org/10.1007/s11042-019-07988-1).
- [46] I. Bakkouri and K. Afdel, "Multi-scale CNN based on region proposals for efficient breast abnormality recognition," *Multimedia Tools Appl.*, vol. 78, no. 10, pp. 12939–12960, May 2019, doi: [10.1007/s11042-018-6267-z](https://doi.org/10.1007/s11042-018-6267-z).
- [47] Z. Ji, X. Li, Z. Wang, H. Zhang, N. Yuan, X. Zhang, and I. Ganchev, "CafeNet: A novel multi-scale context aggregation and multi-level foreground enhancement network for polyp segmentation," *Int. J. Imag. Syst. Technol.*, vol. 34, no. 5, Sep. 2024, Art. no. e23183, doi: [10.1002/ima.23183](https://doi.org/10.1002/ima.23183).
- [48] K. Fang, W. Lu, X. Zhou, J. Xu, and K. Mao, "A multitarget interested region extraction method for wrist X-ray images based on optimized AlexNet and two-class combined model," *IEEE Trans. Computat. Social Syst.*, vol. 9, no. 6, pp. 1624–1634, Dec. 2022, doi: [10.1109/TCSS.2021.3132040](https://doi.org/10.1109/TCSS.2021.3132040).
- [49] L. Chen, Y. Fu, L. Gu, C. Yan, T. Harada, and G. Huang, "Frequency-aware feature fusion for dense image prediction," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 10763–10780, Dec. 2024.
- [50] M. Cordts et al., "The cityscapes dataset for semantic urban scene understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016.
- [51] M. Everingham et al., "The pascal visual object classes (voc) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, 2010.
- [52] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li and L. F.-Fei, "ImageNet: A large-scale hierarchical image database," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, Miami, FL, USA, 2009, pp. 248–255, doi: [10.1109/CVPR.2009.5206848](https://doi.org/10.1109/CVPR.2009.5206848).



JI XIAO received the M.S. degree from Xidian University, Xi'an, China, in 2010. She is currently the Deputy Director of the Digital Technology Department, China Satellite Network Digital Technology Company Ltd. Her research interests include computer vision, data processing, and related areas.



LI JIANFANG received the M.S. degree from the Communication University of China, Beijing, China, in 2016. He is currently pursuing the Eng.D. degree with Wuhan University, Wuhan, China. He is also a Technical Architect with China Satellite Network Digital Technology Company Ltd. His research interests include cloud computing, artificial intelligence, next-generation communication technologies, and related areas.



ZHENG JUNYI received the M.S. degree from Singapore Management University, in 2020. He is currently the Product Manager of China Satellite Network Digital Technology Company Ltd. His professional pursuits involve actively engaging in research on AI applications in integrated space-air-ground systems, including computer vision, knowledge graphs, and decision intelligence.



ZHAO PENG is the Chairperson of China Star Network Digital Technology Company Ltd. His professional fields cover wireless communications, signal processing, data analysis, and related areas. He is the author of several monographs, including IP multimedia concepts and services in mobile communications.



PANG YONGHUI received the B.E. degree from the School of Artificial Intelligence and Data Science, Hebei University of Technology, in 2019. He is currently pursuing the M.S. degree with Hebei University of Technology. His research interests include computer vision, machine learning, and deep learning.



LI XIAOCHEN received the M.S. degree from the Department of Computer Science, Johns Hopkins University, USA, in 2018. He is currently an Architect with China Satellite Network Digital Technology Company Ltd. His research interests include computer vision, natural language processing, and algorithm engineering architecture.



HUANG XIANGSHENG received the B.S. degree in material science and the M.S. degree in computer science from Chongqing University, Chongqing, China, in 1998 and 2002, respectively, and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2005. From 2005 to 2010, he was with the Samsung Advanced Institute of Technology. He is currently an Associate Professor with the Institute of Automation, Chinese Academy of Sciences. He has authored or co-authored over 30 papers in international journals and conferences. He holds ten patents. His current research interests include face technology, 3-D imaging and registration, machine learning, wavelet and filter banks, deep reinforcement learning, neural networks, and reinforcement learning methods.



ZHANG CHENGCHUN received the B.E. degree from the School of Computer Science, Hangzhou Dianzi University, Hangzhou, China, in 2010. He is currently the Technical Manager of China Satellite Network Digital Technology Company Ltd., focusing on technological research and development and industrial applications in the field of computer vision. His research interests include image processing algorithm optimization, object detection, intelligent security system development, and related areas.

...